Problem A. Wired Mall

Input file: `standard input`
Output file: `standard output`
Balloon Color: `Orange`

Coach Mohamed Abd-Elwahab wants you to take a balloon, so he asks for your help with this simple problem. The Coach visits a wired Mall where everything increases by 10 coins every month, and he will tell you the price this month and ask you about its price next month.

Input

Only one line containing a number N $(1 \leq N \leq 90)$.

Output

Print the price of this thing next month.

Examples

standard input	standard output
1	11
2	12

Problem B. Guess Number

Input file:	`standard input`
Output file:	`standard output`
Balloon Color:	`Navy Blue`

There is a hidden integer N such that $1 \le N \le 10^9$. Your task is to find this number using an interactive process.

You may guess any integer X ($1 \le X \le 10^9$). For each guess, the judge will respond with one of the following:

- ">" — if $N > X$,
- "<" — if $N < X$,
- "=" — if $N = X$.

When you find the hidden number, output "`! N`" (where N is the hidden number you found) and terminate your program immediately.

If you make more than 30 guesses, you will receive a Wrong Answer verdict.

Interaction Protocol

This is an interactive problem.

- To make a guess, output a line of the form "`? X`" (without quotes) and flush the output stream immediately.
- After each guess, read the judge's response from the standard input.
- When you find N, output "`! N`" and flush the output immediately.

If you do not flush the output after each guess, you may get an "Idleness limit exceeded" verdict.

To flush the output stream, you may use:

- `fflush(stdout);`, `cout.flush();` or `endl` in C++,
- `System.out.flush();` in Java,
- `stdout.flush()` in Python,

Example

standard input	standard output
<	? 11
>	? 6
>	? 9
=	? 10
	! 10

Note

Your final guess ("`! N`") will not be counted among the 30 allowed guesses.

Problem C. Academic Schedule Optimization

Input file:	`standard input`
Output file:	`standard output`
Balloon Color:	Cyan

Qassem is not a fan of coffee, but he is facing a challenge that might require him to drink it. He is a student in the engineering faculty, where many important lectures are scheduled. Each lecture is defined by when it starts, when it ends, and the level of focus it demands.

To join a lecture, *Qassem* needs to have a focus level at least as high as the lecture's requirement. His goal is to attend as many lectures as possible with the least amount of coffee, so he wants to keep his focus level just enough to meet the highest requirement among the lectures he attends.

Qassem cannot attend overlapping lectures — no two lectures he attends can overlap at any single point in time. That is, if one lecture ends at time t, the next lecture must start strictly after t.

Your task is to help *Qassem* plan which lectures to attend to maximize the total number while minimizing his required focus level.

Input

The first line contains an integer T ($1 \le T \le 100$), the number of test cases.

The first line of each test case contains an integer N ($1 \le N \le 10^5$), representing the number of lectures.

The next N lines of each test case describe each lecture with three integers: S, E, C ($1 \le S \le E \le 10^{18}$), ($1 \le C \le 10^{18}$), representing start time, end time, and the focus level required.

It is guaranteed that the sum of N over all test cases doesn't exceed 10^5.

Output

For each test case, output two integers: the maximum number of lectures Qassem can attend and the minimum possible focus level required to attend them.

If there are multiple ways to attend the maximum number of lectures, choose the one that requires the least focus level.

Example

standard input	standard output
2 3 1 3 2 2 4 3 3 5 2 5 1 3 2 5 8 3 7 12 2 15 18 4 14 20 1	1 2 3 2

Problem D. SuperDuper Strings

Input file:	`standard input`
Output file:	`standard output`
Balloon Color:	`Gray`

You are given an array A of n strings and q queries.

For the i-th query, you are given an integer x_i $(1 \le x_i \le n)$ and a string y_i.

Your task is to count how many substrings of A_{x_i} have t_i as their *SuperDuper* string.

A string B is called a *SuperDuper* string of a string S if S can be formed by concatenating one or more copies of B. For example:

- If $S =$ abcabcabc, then "abc" is a *SuperDuper* string of S (since "abc" repeated 3 times equals S).
- If $S =$ abca, then "abc" is **not** a *SuperDuper* string of S.

Input

The first line contains a single integer n $(1 \le n \le 10^5)$ — the number of strings.

The next n lines each contain a string A_i $(1 \le |A_i| \le 10^5)$, where $|A_i|$ is the length of A_i.

The next line contains a single integer q $(1 \le q \le 10^5)$ — the number of queries.

Each of the next q lines contains an integer x $(1 \le x \le n)$ and a string y $(1 \le |y| \le 10^5)$.

It is guaranteed that:

- The total length of all strings A_i does not exceed 10^5.
- The total length of all query strings y does not exceed 10^5.

Output

Output q integers. The i-th integer should be the answer to the i-th query.

Examples

standard input	standard output
3 abc abcabcxyzabc abcabc 3 1 abc 2 abc 3 xyz	1 4 0
4 xyzfoobarfooxyzabckoko abcxyzfoobarfooxyzkoko foobarfooxyzabcabckoko kokokokokoko 10 2 foobarfoo 3 xyz 2 koko 3 koko 3 abc 2 koko 4 koko 2 koko 2 xyz 3 foobarfoo	1 1 1 1 3 1 9 1 2 1

Note

Suppose A_{x_i} = abcabcxyzabc and y_i = abc. Then the substrings of A_{x_i} that have a *SuperDuper* string equal to y_i are: {"abc", "abcabc", "abc", "abc"}.

Note: A string S may have more than one *SuperDuper* string.